

Quiz Questions:

- **Question 1:** What is the purpose of the `for` loop in Python?
 - a) To check a condition and execute a block of code if it's true.
 - b) To repeatedly execute a block of code for each element in a sequence.
 - c) To perform arithmetic operations.
 - d) To define a function.
- **Question 2:** How do you iterate over a range of numbers in a `for` loop?
 - a) Using the `enumerate()` function.
 - b) Using the `range()` function.
 - c) Using the `while` loop.
 - d) Using the `list()` function.
- **Question 3:** When does a `while` loop stop executing?
 - a) When the loop variable reaches the end of the sequence.
 - b) When the loop condition becomes false.
 - c) When the loop variable is equal to a specific value.
 - d) When the loop is explicitly terminated using `break`.
- **Question 4:** What does the `while` loop syntax look like in Python?
 - a)
`for variable in iterable:`
 - b)
`while condition:`
 - c)
`if condition:`
 - d)
`loop while condition:`

Coding Exercise:

Exercise 1: Sum of Squares

Write a Python program that calculates and prints the sum of the squares of numbers from 1 to 5 using a

`for` loop.

Exercise 2: Countdown

Write a Python program that uses a

`while` loop to print a countdown from 5 to 1.

Exercise 3: Multiplication Table with Nested For Loop

Write a Python program to print the multiplication table for a user-specified number using a nested for loop.

Exercise 4:

Write a Python program that uses a "for" loop to find the sum of all even numbers between 0 and 10 (inclusive).

Exercise 5:

Calculate the sum of all numbers from 1 to a given number

Exercise 6:

Display numbers from a list using loop

Exercise 7:

Display numbers from -10 to -1 using for loop

Exercise 8:

Write a program to display all prime numbers within a range